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An empirical investigation into supply chain vulnerability

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Abstract

A growing number of academicians and practitioners have put supply chain risks on their agendas, particularly triggered by a recent series of catastrophic events that have disrupted economies and supply chains around the globe. Given the increasing awareness of this important topic, the purpose of this research was to study supply chain risks in more detail and to investigate the relationship between supply chain vulnerability and supply chain risk. Responses from 760 executives from firms operating in Germany reveal that supply chain characteristics such as a firm's dependence on certain customers and suppliers, the degree of single sourcing, or reliance on global supply sources are relevant for a firm's exposure to supply chain risk. Overall, this research represents the first large-scale investigation of this important relationship and provides a finer understanding of the antecedents of supply chain vulnerability.
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1. Introduction

In the past five years, a subject in the field of supply chain management research has gained considerable attention by both academicians and practitioners: supply chain risk management. The high regard that firms and researchers are paying to the topic is notably triggered by the frequency and intensity of catastrophes, disasters, and crises that seem to have increased on a global scale (Coleman, 2006; Helferich and Cook, 2002; Munich Re, 2006). The terrorist attacks on the World Trade Center on September 11, 2001, the SARS epidemic in South-East Asia in 2003, and the natural disaster of Hurricane Katrina in 2005 are three exemplary disasters and violent reminders that firms and their global supply chains operate in an unpredictable and increasingly uncertain environment. But not only these prominent "macro" events lead to costly supply chain disruptions. There is also a substantial body of recent literature that reports on events on the "supply chain level", which resulted in serious problems for the involved firms (e.g., Latour, 2001; Norrman and Jansson, 2004; Sheffi, 2005). In Germany for instance, Robert Bosch GmbH, currently the world's largest auto-parts supplier, faced a severe supply chain disruption in January 2005. At that time, Bosch failed to detect a defect in the Teflon coating on a 1.5 cm small socket (worth only a few Eurocents) that goes into some of its diesel injection pumps supplied to automotive OEMs such as Audi, BMW, and DaimlerChrysler. What makes this example noteworthy is that the socket was not produced by Bosch but by its US supplier Federal Mogul which in turn sources the Teflon from DuPont. Downstream the supply chain, the defect Teflon coating inside the diesel injection pumps resulted in a standstill of some of the OEMs' assembly lines, a product recall of several thousand cars, consequential costs in the three digit million Euro area, and a negative impact on the brand image of Bosch and the involved OEMs (Wagner, 2006).

Based on a larger sample, Hendricks and Singhal (2003, 2005a, b) underscore the relevance of supply chain disruptions by showing empirically that these events have a significant negative impact on shareholder value (2003, 2005a) and on operating performance (i.e., sales, operating income, return on assets) (2005b). In essence, it can be concluded from their research that capital markets severely penalise supply chain disruptions and attach high value to

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supply chain robustness. However, having a look at the current situation in business practice, the prevalent supply chain designs do not reflect this call for more robustness. In contrast, it seems that modern supply chains have become more prone to disruptions. Christopher and Lee (2004, p. 388) for instance argue that "the vulnerability of supply chains to disturbance or disruption has increased". This is due to a combination of several factors and trends. In recent years, almost all industries have witnessed fiercer competition and accelerated globalisation of markets. This resulted in a massive pressure to make intrafirm and interfirm business processes more efficient and/or more responsive, for instance, by means of outsourcing and offshoring large portions of manufacturing and R&D activities, sourcing in low-cost countries, reducing inventories and slack, streamlining the supply base, and collaborating more intensively with other supply chain actors (Fisher, 1997; Hult et al., 2004). Often, the consequences of such initiatives are an increase in interfirm dependence as well as longer and more complex supply chain setups with globe-spanning operations which ultimately exacerbate the vulnerability of supply chains to unexpected events (Christopher and Peck, 2004; Harland et al., 2003; Hendricks and Singhal, 2005a; Tang, 2006a, b). Thus, some scholars presume that modern supply chain management initiatives have great potential to make operations leaner and/or more agile in a stable environment but simultaneously amplify the fragility of supply chains, "that is, less able to deal with shocks and disruptions that can have a significant, if not catastrophic, impact on the firm" (Zsidisin et al., 2005b, p. 46). This is supported by findings of organisational scientists showing that due to their increasingly complex and technology-oriented processes, organisations are becoming more prone to disruptions (e.g., Grabowski and Roberts, 1997; Lin et al., 2006). This reasoning traces back to the socio-technological perspective of Perrow (1984) who proposed the Normal Accident Theory that holds the core hypothesis that in complex, tightly coupled technological systems accidents become inevitable, i.e. normal. Drawing from this theory, one could assume that the more complex the interactions and the tighter coupled the supply chain, the more prone the supply chain is to unexpected, untoward events.

Given this relevance, numerous proposals for best practices as well as guidelines for risk mitigation and business continuity planning that aim at ultimately creating secure, robust, and/or resilient supply chains have been published recently (e.g., Chopra and Sodhi, 2004; Lee and Wolfe, 2003; Martha and Subbakrishna, 2002; Rice and Caniato, 2003; Zsidisin et al., 2005b). In spite of their practical value, the vast majority of these contributions is of normative nature, anecdotal or case study-based. Results of large-scale empirical research on supply chain risk issues are scarce. In particular, current knowledge about the mechanisms and conditions that determine the vulnerability of supply chains and about the interaction of supply chain vulnerability and supply chain disruptions is quite limited. To the authors' knowledge, only the event study-based contribution of Papadakis (2006) has attempted to shed empirically substantiated light on the vulnerability of supply chains by analysing and comparing the reaction of the stock performance of firms with make-to-order (MTO) supply chains and firms with make-to-forecast (MTF) supply chains after the 1999 earthquake in Taiwan.

A better understanding of how supply chain design effects supply chain risk exposure would enable supply chain managers to structure their supply chains in a way that they are in line with the firms' willingness to take risks and hence would enable better supply chain design decision making. For instance, a firm that seeks to take advantage of economies of scale in its inbound supply chain by pursuing a single sourcing strategy concurrently might suffer from an increased negative impact of supply-side disruptions (e.g., supplier defaults). While this relationship might seem to be intuitively obvious at the first sight, it lacks empirically substantiated quantification.

By means of a large-scale survey, the objective of this research is to investigate the relationship between a set of supply chain characteristics—which are supposed to provoke the vulnerability of supply chains—and the impact on the performance from three classes of supply chain disruptions.

The rest of the article is organised as follows: In Section 2, we introduce the conceptual background of this study and put forward our model's underlying notions of the terms "supply chain risk", "supply chain disruption", and "supply chain vulnerability". Moreover, the model and its hypotheses are developed by elaborating the relationship between supply chain vulnerability and supply chain risks. Section 3 describes the empirical study and the methodology used to test the hypothesised relationships. Section 4 presents the results of the study. Finally, Section 5 discusses the findings and Section 6 elaborates on the implications for managerial practice and future research.

2. Nomenclature, theory, and hypotheses

Recently, several publications have advanced the conceptual clarity of the terms used in the domain of supply chain risk management—yet, there is still no commonly agreed nomenclature. Therefore, the purpose of the following three subsections is to outline a consistent nomenclature, which is the basis of the research model delineated in the fourth subsection. However, it has to be pointed out that other nomenclatures exist in the pertinent literature.

In essence, we distinguish four interrelated terms: Supply chain risk, supply chain disruption, supply chain risk source, and supply chain vulnerability. In the following, these terms will be briefly derived from the pertinent literature and discussed.

2.1. Supply chain risk

It is crucial for any study dealing with risk to define the term appropriately since it is an elusive construct with a variety of meanings, measurements, and interpretations depending on the field of research (Baird and Thomas, 1990; Jemison, 1987). There is a large body of literature on risk in such fields as decision theory (e.g., Arrow, 1965), insurance (e.g., Knight, 1921), finance (e.g., Markowitz, 1952), marketing (e.g., Cox, 1967), management (e.g., March and Shapira, 1987), and psychology (e.g., Slovic, 1987).

In the field of supply chain management, several publications have addressed the question of how to define supply chain risk. Here—but this is also valid for the discussion on risk in general—two differing meanings can be distinguished. There is a persistent tension between risk as purely danger and risk as both danger and opportunity (Mitchell, 1995). According to classical decision theory as well as to common practice in many other fields of business research such as finance, the fluctuations around the expected value of a performance measure are used as proxy for risk, i.e. risk is equated with variance and consequently covers both a "downside" and an "upside" potential. Following these considerations and by analogy to the definition of March and Shapira (1987, p. 1404)—risk is the "variation in the distribution of possible outcomes, their likelihoods, and their subjective values"—Jüttner et al. (2003, p. 200) define supply chain risk as a "variation in the distribution of possible supply chain outcomes, their likelihood, and their subjective value".

In contrast, in most dictionaries as well as in the field of insurance, risk is viewed as the chance of injury, damage, or loss (Webster, 1983). The notion that risk inherits primarily negative consequences corresponds to the common human perception. For instance, March and Shapira (1987) empirically examine how managers perceive risk and react to it. They find that the majority tends to overrate the "downside" potential of risk. Several scholars in the supply chain management and supply management field have adopted this view. Harland et al. (2003, p. 52) for instance, discuss several definitions and conclude that supply chain risk is associated with the "chance of danger, damage, loss, injury or any other undesired consequences".

For the purpose of this study and considering the impact of recent catastrophes on supply chains, we find that the latter notion of risk as purely negative is the one that corresponds best to supply chain business reality. For this reason, we will consider neither "happy disasters" nor the situation where managers intentionally "gamble" on risk. Our model considers risk as the negative deviation from the expected value of a certain performance measure, resulting in negative consequences for the focal firm. Hence, risk is equated with the detriment of a supply chain disruption, i.e. the realised harm or loss.

2.2. Supply chain disruption and supply chain risk sources

For the purpose of this article, a supply chain disruption is an unintended, untoward situation, which leads to supply chain risk. For the affected firms, it is an exceptional and anomalous situation in comparison to every-day business. Depending on its severity other terms might be applied, e.g. glitch, disturbance, or crisis. It is triggered by an underlying disruptive event (or a series of such events). There has been intensive research by organisational scientists on events that adversely affect organisations, how organisational crisis emerge from those events, and how organisations react to them (Kovoor-Misra et al., 2001; Rijpma, 1997). Several similarities can be derived from this research and transferred to the supply chain risk context. For instance, similarly to the triggering event of an organisational crisis, the triggering event of a supply chain disruption is "identifiable according to place, time, and agents" (Shrivastava et al., 1988, p. 288). The disruption is associated with a certain probability of occurrence and characterised by its severity as well as direct and indirect effects (Kleindorfer and Saad, 2005). Since the resulting detriment is usually a function of time, supply chain disruptions involve time pressure implying that decisions for mitigation must be made swiftly (Hermann, 1963).

Supply chain disruptions can materialise from various areas internal and external to a supply chain. Consequently, their nature can be highly divergent. For instance, the financial default of a supplier and a natural disaster destroying production capacity are situations with completely different attributes (e.g., incubation period, probability, predictability and precursory signs, and severity) and therefore entail different effects on the supply chain. Addressing this issue and attempting to circumscribe supply chain disruptions (viz. the demarcation of supply chain risks from other business risk), many scholars have proposed classifications in the form of typologies and/or taxonomies of risks (e.g., Chopra and Sodhi, 2004; Christopher and Peck, 2004; Hallikas et al., 2004; Jüttner, 2005; Jüttner et al., 2003; Norrman and Lindroth, 2004; Spekman and Davis, 2004; Svensson, 2000). According to Bailey (1994) typologies are primarily conceptual while taxonomies are empirical. The derived classes of supply chain disruptions are often labelled "supply chain risk sources". For instance, Svensson (2000) names two classes (quantitative and qualitative), Jüttner (2005) proposes three classes (supply, demand, and environmental), and Chopra and Sodhi (2004) delineate nine of them (disruptions, delays, systems, forecast, intellectual property, procurement, receivables, inventory, and capacity). For brevity, we call a negative deviation from the expected value of a performance measure (resulting in negative consequences for the focal firm) for example a "supply-side risk" when this deviation is the result of a supply chain disruption that emerged from the supply-side risk source such as the aforementioned supplier quality situation of Bosch.

2.3. Supply chain vulnerability and its drivers

While a supply chain disruption is the situation that leads to the occurrence of risk, it is not the sole determinant of the final result. It seems consequential that also the susceptibility of the supply chain to the harm of this situation is of significant relevance. This leads to the concept of supply chain vulnerability. The example of Bosch and the defective Teflon coating depicts that strong reliance on external sources and tight coupling between supply chain partners heighten the susceptibility to disruptive events.

Although the literature offers numerous approaches to the construct "supply chain vulnerability", Peck (2005) still appraises its conceptual basis as immature. Christopher and Peck (2004, p. 3) define supply chain vulnerability as "an exposure to serious disturbance". Svensson (2000, 2002, 2004a, b) published several contributions that shed light on this construct. He distinguishes between atomistic vulnerability (of a part of the supply chain) and holistic vulnerability (across the entire supply chain). In the literature on natural hazard and crisis management, vulnerability has been defined as a person's (or a group's) capacity to anticipate, cope with, resist, and recover from the impact of a natural hazard (Blaikie et al., 1994). In the context of maritime supply chains, Barnes and Oloruntoba (2005, p. 519) describe vulnerability as "a susceptibility or predisposition to (…) loss because of existing organisational or functional practices or conditions".

In this article, this latter definition is applied and the atomistic perspective (supply chain vulnerability on the individual firm level) is taken. We posit that supply chain vulnerability is a function of certain supply chain characteristics and that the loss a firm incurs is a result of its supply chain vulnerability to a given supply chain disruption.

Certainly, supply chain vulnerability is a multifaceted, multidimensional construct. In the literature, several factors are repeatedly cited as antecedents of supply chain vulnerability such as single sourcing or global sourcing. However, to our knowledge, so far, there is no published, empirically tested operationalisation of supply chain vulnerability. Thus, we will single out a small set of these supply chain vulnerability drivers and empirically investigate how these relate to supply chain risk.

2.4. Model and hypotheses

Based on the nomenclature outlined in the previous subsections, previous research, and selected theories, we will derive our research model and hypotheses in the following. The basic premise underlying the model is that certain supply chain characteristics are drivers of supply chain vulnerability and that there is a relationship between these drivers and the negative impact experienced by firms as a result of supply chain disruptions.

2.4.1. Supply chain risk sources

As discussed in Section 2.2, various different classifications of supply chain disruptions exist. The model of the present research incorporates three supply chain risk sources: demand-side, supply-side, and catastrophic. Certainly, these three classes are not exhaustive and do not cover all risk events inherent in supply chains, e.g. disruptions deriving from internal operations or disruptions caused by legislative changes. However, for our investigation, we chose these three risk sources because demand- and supply-side risks belong unquestionable to the most important issues in supply chain risk management and are therefore of high relevance for practitioners (Kleindorfer and van Wassenhove, 2004) and catastrophic risks are currently very topical and receive ample attention (Sheffi, 2005).

2.4.1.1. Demand-side risk. Demand-side risks result from disruptions emerging from downstream supply chain operations (Jüttner, 2005). This includes on the one hand disruptions in the physical distribution of products to the end-customer with particular issues being transportation operations (e.g., a truck driver strike) (McKinnon, 2006) and the distribution network (e.g., a fire in a warehouse). On the other hand, demand-side risks can originate from the uncertainty surrounding the random demands of the customers (Nagurney et al., 2005). Disruptions occur here from a mismatch between a company's projections and actual demand as well as from poor supply chain coordination. Consequence of which are costly shortages, obsolescence, and inefficient capacity utilisation. An important issue in this context, affecting forecast quality and therefore demand-side disruptions, is the bullwhip effect, which is characterised by an amplification of demand volatility in the upstream direction of the supply chain. Lee et al. (1997) analysed this detrimental effect and identified delayed and distorted information, sales promotions, order batching, price fluctuations and rationing, or shortage gaming as major causes. Other factors intensifying the bullwhip effect are over-reactions, unnecessary interventions, second guessing, and mistrust (Christopher and Lee, 2004). Although demand-side risk management could be labelled the "bread-and-butter" discipline of supply chain management, it still presents a major risk source for many firms. Spekman and Davis (2004) cite the example of Cisco Systems Inc. that wrote off US$ 2.5 billion in inventory in 2001 due to a lack of communication among its downstream supply chain partners.

2.4.1.2. Supply-side risk. Purchasing organisations are exposed to numerous risks associated with their suppliers and their supply network. Kraljic (1983) was among the first who emphasised that firms should proactively assess and manage the uncertainties in their supplier portfolio in order to guard against costly supply disruptions. Moreover, this is highlighted by the trend towards more reliance on external sources (Giunipero and Eltantawy, 2004).

Supply-side risks reside in purchasing, supplier activities, and supplier relationships. These include supplier business risks, production capacity constraints on the supply market, quality problems, technological changes, and product design changes (Zsidisin et al., 2000).

Supplier business risks relate to the various events that affect the continuity of the supplier and result in the temporary or permanent perturbation or termination of the buyer–supplier relationship. This concerns particularly the threat of financial instability of suppliers and the consequences of supplier default, insolvency, or bankruptcy (Wagner and Johnson, 2004). Another type of disruption occurs when a supplier is vertically integrated by a direct competitor of the customer firm, forcing the termination of the relationship (Chopra and Sodhi, 2004). In cooperative settings, opportunistic behaviour from suppliers has also been reported in the literature as a source of supply risk (Smeltzer and Siferd, 1998; Spekman and Davis, 2004; Williamson, 1985). Particularly organisational lock-in is a threat where a purchasing organisation is so dependent on a certain supplier that it has only a limited room for manoeuvring.

Capacity constraints or shortages as well as poor logistics performance (delivery reliability) derive from unsolved problems in the supplier's production and operations management (Lee and Billington, 1993). The bullwhip effect cited in the previous paragraph plays a role here as well and has to be tackled by the suppliers. Furthermore, poor quality in the purchased products or services is a significant risk and can have a cascading effect through the supply chain to the final customer (Zsidisin et al., 2000).

Finally, the inability of suppliers to adapt to technological or product design changes may have detrimental effects on the customer's costs and competitiveness (Zsidisin and Ellram, 2003).

2.4.1.3. Catastrophic risk. This class subsumes supply chain disruptions that, when they materialise, have a severe impact in terms of magnitude in the area of their occurrence. This refers to natural hazards (force majeure), socio-political instability, civil unrest, economic disruptions, and terrorist attacks (Kleindorfer and Saad, 2005; Martha and Subbakrishna, 2002).

In many regions of the world, natural hazards such as tsunamis, droughts, earthquakes, hurricanes, and floods are a constant threat to societies in general and to firms in particular (Helferich and Cook, 2002). The negative consequences on supply chains are obvious since production facilities and transportation are highly vulnerable to natural disasters. Due to the globalisation of markets and a surge in globe-spanning supply chain operations, local catastrophes increasingly have indirect global repercussions. For instance, the 3PL industry—an important element of the German industry landscape—has experienced considerable problems in the aftermath of September 11, 2001. Terrorism is a special topic, which has been discussed extensively in the supply chain context. In the current state of the world, terrorism is a threat to global supply chains that has to be considered (Sheffi, 2005). These attacks can impact supply chains either directly or indirectly. In addition, there are indirect consequences of terrorism that are not caused by an attack itself but by the reaction of governments and markets. For example Ford, Toyota, and DaimlerChrysler experienced massive disruptions to the flow of materials into their North-American assembly plants within a few days after the terrorist attack of September 11 (Sheffi, 2001).

2.4.2. Supply chain vulnerability drivers

Several publications mention how certain supply chain characteristics might increase or decrease the vulnerability of the supply chain. We found assumptions that supply chain vulnerability is increased by customer dependence (Hallikas et al., 2005; Svensson, 2004a), supplier dependence (Giunipero and Eltantawy, 2004; Hendricks and Singhal, 2005a; Jüttner, 2005; Spekman and Davis, 2004; Svensson, 2004a), supplier concentration (Norrman and Jansson, 2004; Tang, 2006a; Zsidisin et al., 2000), single sourcing (Hendricks and Singhal, 2005a; Zsidisin et al., 2004), and global sourcing (Jüttner, 2005; Kraljic, 1983; Peck, 2005, 2006; Seshadri and Subrahmanyam, 2005).

2.4.2.1. Customer dependence and supplier dependence. Dependence is a property of buyer–supplier relationships (Pfeffer and Salancik, 1978). *Supplier dependence* has been viewed as the extent to which an organisation sources inputs from a supplier (or more suppliers) for which there are few alternative sources (Hallikas et al., 2005; Hibbard et al., 2001). In such a setting, the customer firm is vulnerable since it has limited room for manoeuvring while the supplier is dominant and holds power (Bourantas, 1989). In case of a supply-side disturbance, the buying firm will experience significant problems in substituting the supply by a contingency source. The severity of the disruption is particularly amplified by the criticality of the purchased item (Giunipero and Eltantawy, 2004).

Customer dependence is similar to supplier dependence but transferred to a downstream relationship. The focal firm is dependent on some of its customers, e.g. because of a high sales volume. In case of a disruption, the focal firm might be forced to bear a large portion of the impact because of the dominant customers.

2.4.2.2. Supplier concentration and single sourcing. *Supplier concentration* is a scenario where the buying firm only has a small number of suppliers. There has been a recent trend to reduce the supply base and at the same time to establish closer relationships with the suppliers in the smaller supply base. Various benefits of supply base reduction efforts are cited in the literature such as improved product (Kekre et al., 1995) or relationship quality (Ellram, 1991). However, the degree of relying on the external sources and rationalising the supply base

affects supply chain vulnerability (Cavinato, 2004; Choi and Krause, 2006; Svensson, 2004a). When a firm concentrates its sourcing activities on a small number, it loses the ability to switch to contingency suppliers in disruptive situations. Moreover, the buying firm is exposed to a heightened risk of organisational hold-up. *Single sourcing* is an extreme case of supplier concentration.

Recent literature makes a case for not reducing the supply base without deliberately considering the consequences in terms of risk exposure arising from it (Christopher and Peck, 2004; Elkins et al., 2005; Giunipero and Eltantawy, 2004). A common ex-ante strategy to safeguard against the consequences of a sudden shortfall in supply is the diversification of the supply base (Anupindi and Akella, 1993; Treleven and Schweikhart, 1988). The rationale is to install redundancy by developing contingency supply sources in order to decrease the vulnerability to supply-side disruptions (Sheffi, 2005). Having multiple competing suppliers, the buying firm can diversify order quantities and hedge against the sudden demise of a single supplier.

2.4.2.3. Global sourcing. The benefits of global sourcing depend heavily on parameters such as the geographic location of the suppliers, the product purchased, or the mode of transportation. However, in comparison to sourcing from local markets, global sourcing is usually associated with increased uncertainty as well as poorer transparency and visibility. In addition, complicating factors that have to be tackled are longer lead-times due to long routes of transportation, reliance on critical infrastructures (ports, communication systems), taxes, duties, and fluctuations of exchange rates (Goetschalckx et al., 2002). In general, global sourcing contributes to the structural complexity of the supply chain (Hendricks and Singhal, 2005a). Here, Normal Accident Theory that was mentioned in the introduction yields explanatory power. This theory argues that the complexity of a system and the coupling of its elements are the two major determinants of system accidents. Hence, given that this concept holds for supply chains, a more complex supply chain is more prone to disruptions. This in turn asks for rigorous control mechanisms and administration (Christopher and Lee, 2004; Perrow, 1984, 1999).

2.4.3. Hypotheses

Based on this analysis and the previous discussions on the phenomenon of supply chain risk, we posit that the five drivers of supply chain vulnerability are positively related to the risk emerging from the three supply chain risk sources. Therefore, we formulate the following three (aggregated) hypotheses:

H1. The higher the drivers of supply chain vulnerability, the higher the level of demand-side risk a firm faces.

H2. The higher the drivers of supply chain vulnerability, the higher the level of supply-side risk a firm faces.

H3. The higher the drivers of supply chain vulnerability, the higher the level of catastrophic risk a firm faces.

3. Methods

3.1. Data collection and sample

Data were collected through a cross-sectional survey administered in Germany to a sample of 4946 top-level executives in logistics and supply chain management. The mailing and two follow-ups generated 760 usable responses, yielding a relatively high response rate of about 15.4%, considering the time constraints of top-level executives (Baldauf et al., 1999; Tomaskovic-Devey et al., 1994). Non-response bias was assessed on the notion that later respondents would be more like non-respondents (Armstrong and Overton, 1977). For all questionnaire items, the responses of later respondents were compared to those of earlier. This comparison indicated absence of non-response bias.

The sample covered industrial (71.7% of the sample), service (19.5%) and trade (8.8%) firms. The firms' annual sales ranged from less than US$ 10 million to US$ 90 billion (mean US$ 60.3 million), and the number of employees from less than 100 to 430,000 (mean 2913), thus yielding a heterogeneous sample. Given the range and size of the firms studied and the diversity of industries, there was no prima facie reason to expect any systematic bias in the results. Most of the respondents hold executive positions in logistics and supply chain management (37.5%), or were C-level executives and owners (23.8%). On average, the respondents have worked in their position for 7.0 years and have been with the firm for 10.9 years. A more detailed breakdown of the sample and informants can be found in Table 1.

3.2. Questionnaire and measure development

As indicated earlier, most previous studies on supply chain risk management are based on anecdotal evidence or case studies. Therefore, item batteries with formal scale characteristics that could have been used for this study were not available. New measures and a fully standardised survey instrument were developed in several stages.

First, establishing the content validity of the survey instrument, we based the supply chain risk measures on an initial pool of scale items that had been generated through an extensive review of the academic and practitioner literature on risk management and supply chain management. Operational definitions for the risk measures and construct items were derived from these sources. A preliminary questionnaire was drafted. Second, the scale items included in the questionnaire, their relevance, their wording and directions, and the format of the questionnaire were refined on the basis of comments from practitioners and researchers. Third, to further refine the survey instrument, it was pre-tested through interviews with supply chain management executives from a small

Table 1
Sample characteristics

	Percent of total sample (%)	
Sector and industry		
Industry sector		71.5
Automotive	11.2	
Electro/electronics	10.1	
Machinery	9.5	
Chemicals and pharmaceutical	8.4	
Information technology	6.6	
Materials and metal production	6.2	
Food	5.5	
Paper, pulp, and printing	4.2	
Construction	3.0	
Consumer goods	2.5	
Aerospace and defence	2.1	
Medical devices	1.3	
Other industry	0.9	
Service sector		19.5
Logistics services	17.1	
Other services	2.4	
Trade sector		8.8
Sales (in US$)		
Less than 10 million		14.9
10 million—under 50 million		23.9
50 million—under 100 million		16.3
100 million—under 250 million		14.7
250 million—under 500 million		8.7
500 million—under 1 billion		6.7
1 billion—under 10 billion		7.2
10 billion and more		5.0
n.a.		2.5
Number of employees		
Less than 100		21.4
100–499		42.2
500–999		11.6
1000–4999		15.3
5000–9999		2.8
10,000 and more		3.7
n.a.		3.0
Position of informant		
Logistics/supply chain management		37.5
C-level executives or owners		23.8
Purchasing/procurement		15.0
Production/operations		13.9
Sales, distribution, and service		3.6
Other senior management		3.4
Accounting/finance		2.1
n.a.		0.6

number of firms. The executives' comments were incorporated in the final version of the questionnaire.

Respondents were asked to indicate how their firms were negatively affected (supply chain risk) by supply chain disruptions and how their supply chain design was structured (i.e., the drivers of supply chain vulnerability). Five-point Likert-type items were used to operationalise all constructs. All items were scored so that higher numbers reflect increases in the underlying constructs. Translations of the individual scale items and response cues for each measure are listed in Appendix A.

The *demand-side risk* measure consists of two items that capture the detrimental effects of the interaction (or lack of interaction) with customers to be served through the downstream supply chain and volatility on the market (Hendricks and Singhal, 2005a; Jüttner, 2005; Towill, 2005). Likewise, the *supply-side risk* measure consists of five items that capture the detriments stemming from disruptions on the upstream side of the supply chain such as the supply market conditions and the performance of suppliers (Chopra and Sodhi, 2004; Zsidisin, 2003; Zsidisin and Ellram, 2003; Zsidisin et al., 2000, 2005a). For the *catastrophic risk* measure, we generated a four-item scale that captures risks that originate from terrorist acts, socio-political crises, natural disasters, and epidemics (Chopra and Sodhi, 2004; Kleindorfer and Saad, 2005; Martha and Subbakrishna, 2002; Sheffi, 2001).

As mentioned above, a supply chain vulnerability construct has not yet been operationalised in the literature. With the literature review as our point of departure, we measured factors that are supposed to increase supply chain vulnerability (*supply chain vulnerability drivers*). Five single items were used, related to the dependence of the firm on some customers and suppliers, the concentration of the firm's supplier base, the firm's application of single sourcing, and the firm's use of a global supplier base (global sourcing). Along the lines of Rossiter (2002, p. 331) who remarks that a "concrete singular object to be rated in terms of a concrete attribute needs only a single-item scale", we decided to use single items for the supply chain vulnerability drivers because we measured relatively simple or single-faceted circumstances. Other advantages, such as simplicity and brevity of the survey instrument, are obvious.

As firm size can be a potential source of variance that confounds research findings, we included firm size as control variable in our analyses when we tested the effect of supply chain vulnerability drivers on supply chain risk. This control variable was measured by a single item asking respondents for the number of employees in their firm.

Traditional psychometric approaches were used to evaluate the multi-item scales' reliability and validity (Mentzer and Flint, 1997; Nunnally and Bernstein, 1994). They included correlation analysis, principal component factor analysis using varimax as the method of rotation, and reliability evaluation. Factor analysis results and reliabilities are included in Appendix A. Reliabilities for all dependent variables were evaluated via item-to-total correlations and Cronbach's alpha coefficient (Malhotra, 2004; Nunnally and Bernstein, 1994). All item-to-total correlations are above 0.5, i.e. have values greater than 0.35, a threshold that indicates that an item should be deleted from the scale. Cronbach's alpha coefficients range from 0.724 to 0.854. As a rule, coefficients above 0.6 are professionally acceptable, in particular for new scales. All items meet established standards for convergent validity, i.e. all items load on unique components with factor loadings larger than 0.5. In summary, the evidence provided in these analyses suggests that the measures included in this study possess sufficient reliability and

Table 2
Summary statistics and correlation matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Customer dependence	n.a.								
(2) Supplier dependence	0.25***	n.a.							
(3) Supplier concentration	0.10**	0.32***	n.a.						
(4) Single sourcing	0.13***	0.40***	0.31***	n.a.					
(5) Global sourcing	−0.15***	0.07*	-0.05	0.10**	n.a.				
(6) Demand-side risk	0.20***	0.18***	0.11**	0.12***	0.05	n.a.			
(7) Supply-side risk	0.04	0.26***	0.04	0.22***	0.21***	0.40***	n.a.		
(8) Catastrophic risk	0.02	−0.04	0.02	0.03	0.16***	0.13***	0.31***	n.a.	
(9) Firm size	−0.03	−0.00	−0.04	0.01	0.08*	−0.01	−0.00	0.02	n.a.
Mean	3.35	3.03	2.80	2.41	3.00	3.25	2.47	1.55	2.913
Standard deviation	1.27	1.23	1.11	1.14	1.37	0.99	0.77	0.73	19.104

***Significant at the 0.001 level.
**Significant at the 0.01 level.
*Significant at the 0.05 level.

validity to proceed with hypothesis testing. For hypothesis testing analysis, summated composites of the multi-item measures were calculated. Correlations of constructs and descriptive statistics are summarised in Table 2.

4. Results

In order to test our research hypotheses, three ordinary least square (OLS) regression models were estimated. Table 3 shows the model summaries as well as the standardised parameter estimates and significance levels. For each model, the assumptions underlying OLS regression (in particular: multicollinearity, autocorrelation, and heteroscedasticity) were checked and revealed no indication for the inappropriateness of the chosen method.

First, the outlined drivers of supply chain vulnerability explained 7% of the variance of demand-side risk ($F = 8.58$, $df = 701, 6$). H1 posits a relationship between the drivers of supply chain vulnerability and demand-side risk. This is supported by two factors that significantly increase demand-side risk: strong customer dependence (standardised parameter estimate = 0.17) and strong supplier dependence (0.10). The other independent variables showed no significant effect. Second, the drivers of supply chain vulnerability explained 13% of the variance of supply-side risk ($F = 16.83$, $df = 702, 6$). H2 positing that the identified drivers of vulnerability have a positive effect on supply-side risk was confirmed for supplier dependence (0.23), single sourcing (0.13), and global sourcing (0.18). In the third model, the drivers of supply chain vulnerability explain solely 3% of the variance of catastrophic risks ($F = 3.48$, $df = 701, 6$). There is a highly significant positive relationship of a firm's reliance on a global sourcing network and the degree of catastrophic risk it experiences (0.16). This provides some support for H3. In addition, a very slight but significant negative relationship (−0.09) between strong supplier dependence and catastrophic risks exists, i.e., the more a firm is dependent on a few suppliers, the less it experiences a negative impact from catastrophic risks.

In each of the three estimated models, the control variable firm size showed no significant effect.

5. Discussion and implications

Preliminarily, it is important to remark that the present study does not attempt to answer the question of how supply chain disruptions actually influence supply chain performance or firm performance. Accordingly, the results allow no statement pertaining to the general relevance of supply chain risks. So far, only the important work of Hendricks and Singhal (2003, 2005a, b) yields empirical insights into the relationship between supply chain disruptions and performance. Their results highlight that stock markets severely penalise announcements of supply chain disruptions. Moreover, our research model does not address the causalities underlying supply chain disruptions. The results do not allow drawing conclusions in the sense of "single sourcing causes supply chain disruptions", for instance. In contrast, based on a sample of 760 top-level executives in logistics and supply chain management, the results of this study notably shed light on the relationship between drivers of supply chain vulnerability and supply chain risk, i.e. the detrimental results from a disruption given certain supply chain characteristics. Starting from a literature review, we identified and singled out several characteristics of supply chain design that are supposed to be relevant for a firm's exposure to supply chain disruptions. Hence, the results attempt to explain how these supply chain characteristics increase or decrease the loss or damage a firm experiences from supply chain disruptions disregarding the actual impact to either the supply chain or the involved firms.

A first aspect to be discussed is that in each of the three estimated models the supply chain vulnerability factors explain a rather small portion of the variance in the dependent variables. This is a clear indication that there are other factors, which are relevant for the investigated relationship. With the five elaborated independent variables, the present research was a first attempt to reveal that

Table 3
OLS regression results

	Dependent variables		
	Demand-side risk	Supply-side risk	Catastrophic risk
Control variable			
Firm size	−0.01	−0.02	0.01
Predictor variables			
Customer dependence	0.17***	0.00	0.05
Supplier dependence	0.10*	0.23***	−0.09*
Supplier concentration	0.06	−0.05	0.04
Single sourcing	0.05	0.13***	0.03
Global sourcing	0.07	0.18***	0.16***
Model summary	$F(701, 6) = 8.58$*** $R^2 = 0.07$	$F(702, 6) = 16.83$*** $R^2 = 0.13$	$F(701, 6) = 3.48$** $R^2 = 0.03$

***Significant at the 0.001 level.
**Significant at the 0.01 level.
*Significant at the 0.05 level.

there are significant statistical relationships between supply chain design and supply chain risk. However, it seems to be intuitive that beyond the scope of the estimated models there are more factors that drive supply chain vulnerability. This makes a strong case for the identification of further underlying factors by exploiting theories that yield exploratory power on supply chain vulnerability such as Normal Accident Theory (Perrow, 1984) and High Reliability Theory (e.g., Roberts, 1990).

Concerning the first regression model, the results reveal that customer dependence is positively related to demand-side risk (negative effects from volatile customer demand or information distortion). Thus, firms that are dependent on some customers are exposed to a higher risk of suffering from the detrimental effects of demand volatility and poor downstream information. This is in line with the conceptual grounds delineated in Section 2.4. In addition, demand-side risk is also increased by a dependence on the other side of the supply chain, namely supplier dependence. Being dependent on some suppliers generally implies a lack of switching options and weak negotiation power. Hence, in this case, a firm is less flexible on the supply side and consequently less capable of dealing with volatile demand and poor information from the demand side.

Second, risk derived from supply-side risk sources is elevated by supplier dependence, single sourcing, and global sourcing. Supplier dependence obviously amplifies the threat from poor quality, supply shortages, sudden demise of one of these suppliers, and poor logistics performance. Although this argumentation also applies to single sourcing, the single sourcing approach seems to be less hazardous than general dependence on some suppliers. This might be because single sourcing is usually aligned with a closer relationship (e.g., long-term commitment, open communication, mutual trust, and joint activities) that might absorb some of the supply-side risk. Nevertheless, single sourcing is a purchasing strategy that increases the exposure to supply-side risk. Finally, given the numerous pitfalls associates with global sourcing (e.g., Trent and Monczka, 2002), this approach is in general a lever for supply chain risk (Blackhurst et al., 2005). Moreover, our research shows that global sourcing boosts particularly risk stemming from the upstream supply chain.

Third, when it comes to risk from catastrophic risk sources (natural disasters as well as anthropogenic catastrophes like terror attacks), it has to be taken into consideration that the sample data were collected in Germany, which has so far been a very "calm" place with regard to disasters. Here, it is solely global sourcing that is a significant factor that exposes firms to higher risk from catastrophes. The implementation of a global sourcing strategy stretches the supply chain geographically, which ultimately means more peril points for the information and material flow. The robustness and resilience of regional or national supply chains are usually higher. Surprisingly, the study shows that supplier dependence decreases the risk exposure to catastrophes. An explanation for this finding might be some sort of cooperation in the supply base. This receives support from the recent example of the auto-part supplier Takata Corp. In March 2006, one of Takata's airbag production facilities was destroyed, but its customers (e.g., Honda, GM, and DaimlerChrysler) did not face a disruption because Takata's competitors worked together in order to support their customers, the automotive OEMs (Nussel, 2006).

Overall, the findings from the three models provide support for the hypothesis that supply chain characteristics or design variables influence the exposure of the involved firms to the results from supply chain disruptions. However, the relevance of the defined supply chain vulnerability drivers is rather low since they only marginally explain the variance in supply chain risk.

6. Implications and conclusions

The major objective of this research was to examine the relationship between a selection of supply chain characteristics

and supply chain risk, and provide an empirical investigation of the supply chain vulnerability construct. Building on a thorough examination of the supply chain risk typologies provided by the literature, we compiled and empirically validated constructs for different classes of supply chain risk sources by means of a large-scale survey in Germany.

A couple of managerial implications can be deduced from this study. The findings advocate the consideration of risk aspects in supply chain design. As shown, there are several supply chain design variables that amplify a firm's exposure to risk. In particular, an organisation should certainly try to circumvent customer and supplier dependence. In cases where this is not a valid option, other means of improving the supply chain robustness should be considered. Single sourcing and global sourcing are powerful concepts in a stable environment but have to be evaluated from a risk management perspective. The study gives reason to ask for an acceptable risk-benefit trade-off.

Several areas for future research can be highlighted. First, the data for this survey were collected from firms operating in Germany. Therefore, the results can—if at all—be generalised to firms based in countries with similar geographic, political, and economic characteristics. In particular, as shown by Helferich and Cook (2002), Germany has a low exposure to natural disasters. Therefore, a replication of this survey in other countries with presumably different risk profiles (e.g., Japan or the United States) would be a consequential next step.

Second, the present study only encompasses a selection of major disruptions as well as a variety of supply chain vulnerability driver items. The rather low coefficients of determination make a strong case for the further exploration of supply chain vulnerability. A more detailed operationalisation of supply chain characteristics or design variables which are relevant for supply chain vulnerability together with an investigation of their relationship with supply chain risk would be of high managerial relevance. A deeper knowledge of how supply chain characteristics increase or decrease vulnerability and consequently affect supply chain risk exposure would give managers important information for their decisions on supply chain design.

With the preliminary evidence and perspectives provided in this research, a more extensive investigation of supply chain design variables, supply chain vulnerability, and supply chain risks should present intriguing research questions as well as benefits for managers.

Appendix A

See Table A1.

Table A1
Measures

Items (response cues)	Number of items	Cronbach's alpha	Factor loading	Item-to-total correlation
To what extent has your firm in the past 3 years experienced a negative impact in supply chain management due to … (not at all–to a very large extent)				
Demand side risks	2	0.724		
Unanticipated or very volatile demand			0.879	0.568
Insufficient or distorted information from your customer about orders or demand quantities			0.821	0.568
Supply side risks	5	0.799		
Poor logistics performance of suppliers (e.g., delivery dependability, order fill capacity)			0.826	0.695
Supplier quality problems			0.787	0.687
Sudden demise of a supplier (e.g., due to bankruptcy)			0.725	0.550
Poor logistics performance of logistics service providers			0.684	0.504
Capacity fluctuations or shortages on the supply markets			0.516	0.478
Catastrophic risks	4	0.854		
Political instability, war, civil unrest, or other socio-political crises			0.858	0.756
International terror attacks (e.g., 2005 London or 2004 Madrid terror attacks)			0.817	0.672
Diseases or epidemics (e.g., SARS, foot and mouth disease)			0.806	0.672
Natural disasters (e.g., earthquake, flooding, extreme climate, tsunami)			0.799	0.684
Please evaluate the following statements: (strongly disagree–strongly agree)				
Customer dependence				
Our firm strongly depends on some of its customers	1			
Supplier dependence				
Our firm strongly depends on some of its suppliers	1			
Supplier concentration				
Our firm has concentrated its sourcing activities on a small number of suppliers	1			
Single sourcing				
Our firm frequently pursues single sourcing strategies	1			
Global sourcing				
Our firm relies on a global supplier network (global sourcing)	1			

References

Anupindi, R., Akella, R., 1993. Diversification under supply uncertainty. Management Science 39 (8), 944–963.

Armstrong, J.S., Overton, T.S., 1977. Estimating nonresponse bias in mail surveys. Journal of Marketing Research 14 (3), 396–402.

Arrow, K.J., 1965. Aspects of the Theory of Risk-bearing. Yrjö Hahnsson Foundation, Helsinki.

Bailey, K.D., 1994. Typologies and Taxonomies: An Introduction to Classification Techniques. Sage, Thousand Oaks.

Baird, I.S., Thomas, H., 1990. What is risk anyway? In: Bettis, R.A., Thomas, H. (Eds.), Risk, Strategy, and Management. JAI Press, London, pp. 21–52.

Baldauf, A., Reisinger, H., Moncrief, W.C., 1999. Examining motivations to refuse in industrial mail surveys. Journal of the Market Research Society 41 (3), 345–353.

Barnes, P., Oloruntoba, R., 2005. Assurance of security in maritime supply chains: conceptual issues of vulnerability and crisis management. Journal of International Management 11 (4), 519–540.

Blackhurst, J., Craighead, C.W., Elkins, D., Handfield, R.B., 2005. An empirically derived agenda of critical research issues for managing supply-chain disruptions. International Journal of Production Research 43 (19), 4067–4081.

Blaikie, P., Cannon, T., Davis, I., Wisner, B., 1994. At Risk: Natural Hazards, People's Vulnerability, and Disasters. Routledge, London.

Bourantas, D., 1989. Avoiding dependence on suppliers and distributors. Long Range Planning 22 (3), 140–149.

Cavinato, J.L., 2004. Supply chain logistics risks: from the back room to the board room. International Journal of Physical Distribution & Logistics Management 34 (5), 383–387.

Choi, T.Y., Krause, D.R., 2006. The supply base and its complexity: implications for transaction costs, risks, responsiveness, and innovation. Journal of Operations Management 24 (5), 637–652.

Chopra, S., Sodhi, M.S., 2004. Managing risk to avoid supply-chain breakdown. Sloan Management Review 46 (1), 53–61.

Christopher, M., Lee, H.L., 2004. Mitigating supply chain risk through improved confidence. International Journal of Physical Distribution & Logistics Management 34 (5), 388–396.

Christopher, M., Peck, H., 2004. Building the resilient supply chain. International Journal of Logistics Management 15 (2), 1–13.

Coleman, L., 2006. Frequency of man-made disasters in the 20th century. Journal of Contingencies and Crisis Management 14 (1), 3–11.

Cox, D.F., 1967. Risk Taking and Information Handling in Consumer Behavior. Harvard University Press, Cambridge.

Elkins, D., Handfield, R.B., Blackhurst, J., Craighead, C.W., 2005. 18 ways to guard against disruption. Supply Chain Management Review 9 (1), 46–53.

Ellram, L.M., 1991. A managerial guideline for the development and implementation of purchasing partnerships. International Journal of Purchasing and Materials Management 27 (3), 2–8.

Fisher, M.L., 1997. What is the right supply chain for your product? Harvard Business Review 75 (2), 105–116.

Giunipero, L.C., Eltantawy, R.A., 2004. Securing the upstream supply chain: a risk management approach. International Journal of Physical Distribution & Logistics Management 34 (9), 698–713.

Goetschalckx, M., Vidal, C.J., Dogan, K., 2002. Modeling and design of global logistics systems: a review of integrated strategic and tactical models and design algorithms. European Journal of Operational Research 143 (1), 1–18.

Grabowski, M., Roberts, K.H., 1997. Risk mitigation in large-scale systems: lessons from high reliability organizations. California Management Review 39 (4), 152–162.

Hallikas, J., Karvonen, I., Pulkkinen, U., Virolainen, V.-M., Tuominen, M., 2004. Risk management processes in supplier networks. International Journal of Production Economics 90 (1), 47–58.

Hallikas, J., Puumalainen, K., Vesterinen, T., Virolainen, V.-M., 2005. Risk-based classification of supplier relationships. Journal of Purchasing and Supply Management 11 (2/3), 72–82.

Harland, C., Brenchley, R., Walker, H., 2003. Risk in supply networks. Journal of Purchasing and Supply Management 9 (2), 51–62.

Helferich, O.K., Cook, R.L., 2002. Securing the Supply Chain. Council of Logistics Management, Oak Brook.

Hendricks, K.B., Singhal, V.R., 2003. The effect of supply chain glitches on shareholder wealth. Journal of Operations Management 21 (5), 501–522.

Hendricks, K.B., Singhal, V.R., 2005a. An empirical analysis of the effects of supply chain disruptions on long-run stock price performance and equity risk of the firm. Production and Operations Management 14 (1), 35–52.

Hendricks, K.B., Singhal, V.R., 2005b. Association between supply chain glitches and operating performance. Management Science 51 (5), 695–711.

Hermann, C.F., 1963. Some consequences of crisis which limit the viability of organizations. Administrative Science Quarterly 8 (1), 61–82.

Hibbard, J.D., Kumar, N., Stern, L.W., 2001. Examining the impact of destructive acts in marketing channel relationships. Journal of Marketing Research 38 (1), 45–61.

Hult, G.T.M., Ketchen Jr., D.J., Slater, S.F., 2004. Information processing, knowledge development, and strategic supply chain performance. Academy of Management Journal 47 (2), 241–253.

Jemison, D.B., 1987. Risk and the relationship among strategy, organizational processes, and performance. Management Science 33 (9), 1087–1101.

Jüttner, U., 2005. Supply chain risk management—understanding the business requirements from a practitioner perspective. International Journal of Logistics Management 16 (1), 120–141.

Jüttner, U., Peck, H., Christopher, M., 2003. Supply chain risk management: outlining an agenda for future research. International Journal of Logistics: Research and Applications 6 (4), 197–210.

Kekre, S., Murthi, B.P.S., Srinivasan, K., 1995. Operating decisions, supplier availability and quality: an empirical study. Journal of Operations Management 12 (3/4), 387–396.

Kleindorfer, P.R., Saad, G.H., 2005. Managing disruption risks in supply chains. Production and Operations Management 14 (1), 53–68.

Kleindorfer, P.R., van Wassenhove, L.N., 2004. Managing risk in global supply chains. In: Gatignon, H., Kimberly, J.R., Gunther, R.E. (Eds.), The INSEAD-Wharton Alliance on Globalizing. Cambridge University Press, Cambridge, pp. 288–305.

Knight, F.H., 1921. Risk, Uncertainty and Profit. Hart, Schaffner & Marx, Boston.

Kovoor-Misra, S., Clair, J.A., Bettenhausen, K.L., 2001. Clarifying the attributes of organizational crises. Technological Forecasting and Social Change 67 (1), 77–91.

Kraljic, P., 1983. Purchasing must become supply management. Harvard Business Review 61 (5), 109–117.

Latour, A., 2001. Trial by fire: a blaze in Albuquerque sets off a major crisis for cell-phone giants. Wall Street Journal, January 29.

Lee, H.L., Billington, C., 1993. Material management in decentralized supply chains. Operations Research 41 (5), 835–847.

Lee, H.L., Wolfe, M., 2003. Supply chain security without tears. Supply Chain Management Review 7 (1), 12–20.

Lee, H.L., Padmanabhan, V., Whang, S., 1997. Information distortion in a supply chain: the bullwhip effect. Management Science 43 (4), 546–558.

Lin, Z., Zhao, X., Ismail, K.M., Carley, K.M., 2006. Organizational design and restructuring in response to crises: lessons from computational modeling and real-world cases. Organization Science 17 (5), 598–618.

Malhotra, N.K., 2004. Marketing Research: An Applied Orientation, fourth ed. Prentice-Hall, Upper Saddle River.

March, J.G., Shapira, Z., 1987. Managerial perspectives on risk and risk taking. Management Science 33 (11), 1404–1418.

Markowitz, H., 1952. Portfolio selection. Journal of Finance 7 (1), 77–91.

Martha, J., Subbakrishna, S., 2002. Targeting a just-in-case supply chain for the inevitable next disaster. Supply Chain Management Review 6 (5), 18–23.

McKinnon, A., 2006. Life without trucks: the impact of a temporary disruption of road freight transport on a national economy. Journal of Business Logistics 27 (2), 227–250.

Mentzer, J.T., Flint, D.J., 1997. Validity in logistics research. Journal of Business Logistics 18 (1), 199–216.

Mitchell, V.-W., 1995. Organizational risk perception and reduction: a literature review. British Journal of Management 6 (2), 115–133.

Munich Re, 2006. Annual Review: Natural Catastrophes 2005. Munich Re Publications, Munich.

Nagurney, A., Cruz, J., Dong, J., Zhang, D., 2005. Supply chain networks, electronic commerce, and supply side and demand side risk. European Journal of Operational Research 164 (1), 120–142.

Norrman, A., Jansson, U., 2004. Ericsson's proactive supply chain risk management approach after a serious sub-supplier accident. International Journal of Physical Distribution & Logistics Management 34 (5), 434–456.

Norrman, A., Lindroth, R., 2004. Categorization of supply chain risk and risk management. In: Brindley, C. (Ed.), Supply Chain Risk. Ashgate, Hampshire, pp. 14–27.

Nunnally, J.C., Bernstein, I.H., 1994. Psychometric Theory, third ed. McGraw-Hill, New York.

Nussel, P., 2006. What to do when the plant blows up. Automotive News, August 21.

Papadakis, I.S., 2006. Financial performance of supply chains after disruptions: an event study. Supply Chain Management: An International Journal 11 (1), 25–33.

Peck, H., 2005. Drivers of supply chain vulnerability: an integrated framework. International Journal of Physical Distribution & Logistics Management 35 (4), 210–232.

Peck, H., 2006. Reconciling supply chain vulnerability, risk and supply chain management. International Journal of Logistics: Research and Applications 9 (2), 127–142.

Perrow, C., 1984. Normal Accidents. Basic Books, New York.

Perrow, C., 1999. Organizing to reduce the vulnerabilities of complexity. Journal of Contingencies and Crisis Management 7 (3), 150–155.

Pfeffer, J., Salancik, G.R., 1978. The External Control of Organizations—A Resource Dependence Perspective. Harper and Row, New York.

Rice Jr., J.B., Caniato, F., 2003. Building a resilient and secure supply chain. Supply Chain Management Review 7 (5), 22–30.

Rijpma, J.A., 1997. Complexity, tight-coupling and reliability: connecting normal accidents theory and high reliability theory. Journal of Contingencies and Crisis Management 5 (1), 15–23.

Roberts, K.H., 1990. Some characteristics of one type of high reliability organization. Organization Science 1 (2), 160–177.

Rossiter, J.R., 2002. The C-OAR-SE procedure for scale development in marketing. International Journal of Research in Marketing 19 (4), 305–335.

Seshadri, S., Subrahmanyam, M., 2005. Introduction to the special issue on "Risk Management in Operations". Production and Operations Management 14 (1), 1–4.

Sheffi, Y., 2001. Supply chain management under the threat of international terrorism. International Journal of Logistics Management 12 (2), 1–11.

Sheffi, Y., 2005. The Resilient Enterprise: Overcoming Vulnerability for Competitive Advantage. MIT Press, Cambridge.

Shrivastava, P., Mitroff, I.I., Miller, D., Miglani, A., 1988. Understanding industrial crisis. Journal of Management Studies 25 (4), 285–303.

Slovic, P., 1987. Perception of risk. Science 236 (4799), 280–285.

Smeltzer, L.R., Siferd, S.P., 1998. Proactive supply management: the management of risk. International Journal of Purchasing and Materials Management 34 (1), 38–45.

Spekman, R.E., Davis, E.W., 2004. Risky business: expanding the discussion on risk and the extended enterprise. International Journal of Physical Distribution & Logistics Management 34 (5), 414–433.

Svensson, G., 2000. A conceptual framework for the analysis of vulnerability in supply chains. International Journal of Physical Distribution & Logistics Management 30 (9), 731–749.

Svensson, G., 2002. A conceptual framework of vulnerability in firms' inbound and outbound logistics flows. International Journal of Physical Distribution & Logistics Management 32 (2), 110–134.

Svensson, G., 2004a. Vulnerability in business relationships: the gap between dependence and trust. Journal of Business and Industrial Marketing 19 (7), 469–483.

Svensson, G., 2004b. Key areas, causes and contingency planning of corporate vulnerability in supply chains—a qualitative approach. International Journal of Physical Distribution & Logistics Management 34 (9), 728–748.

Tang, C.S., 2006a. Robust strategies for mitigating supply chain disruptions. International Journal of Logistics: Research and Applications 9 (1), 33–45.

Tang, C.S., 2006b. Perspectives in supply chain risk management. International Journal of Production Economics 103 (2), 451–488.

Tomaskovic-Devey, D., Leiter, J., Thompson, S., 1994. Organizational survey nonresponse. Administrative Science Quarterly 39 (3), 439–457.

Towill, D.R., 2005. The impact of business policy on bullwhip induced risk in supply chain management. International Journal of Physical Distribution & Logistics Management 35 (8), 555–575.

Treleven, M., Schweikhart, S.B., 1988. A risk/benefit analysis of sourcing strategies: single versus multiple sourcing. Journal of Operations Management 7 (4), 93–114.

Trent, R.J., Monczka, R.M., 2002. Pursuing competitive advantage through integrated global sourcing. Academy of Management Executive 16 (2), 66–80.

Wagner, S.M., 2006. Die größte Gefahr ist der Kunde. Harvard Business Manager 28 (5), 12–13.

Wagner, S.M., Johnson, J.L., 2004. Configuring and managing strategic supplier portfolios. Industrial Marketing Management 33 (8), 717–730.

Webster, N., 1983. Webster's New Twentieth Century Dictionary, second ed. Simon & Schuster, New York.

Williamson, O.E., 1985. The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting. The Free Press, New York.

Zsidisin, G.A., 2003. A grounded definition of supply risk. Journal of Purchasing and Supply Management 9 (5/6), 217–224.

Zsidisin, G.A., Ellram, L.M., 2003. An agency theory investigation of supply risk management. Journal of Supply Chain Management 39 (3), 15–27.

Zsidisin, G.A., Panelli, A., Upton, R., 2000. Purchasing organization involvements in risk assessment, contingency plans, and risk management: an exploratory study. Supply Chain Management: An International Journal 5 (4), 187–197.

Zsidisin, G.A., Ellram, L.M., Carter, J.R., Cavinato, J.L., 2004. An analysis of supply risk assessment techniques. International Journal of Physical Distribution & Logistics Management 34 (5), 397–413.

Zsidisin, G.A., Melnyk, S.A., Ragatz, G.L., 2005a. An institutional theory perspective of business continuity planning for purchasing and supply management. International Journal of Production Research 43 (16), 3401–3420.

Zsidisin, G.A., Ragatz, G.L., Melnyk, S.A., 2005b. The dark side of supply chain management. Supply Chain Management Review 9 (2), 46–52.